	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	1172 - fwd_kernel_half	(16, 8, 1)x(32, 1, 1)	85.44 ms	164,052,065	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.92 Ghz	[24911] python3.12	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

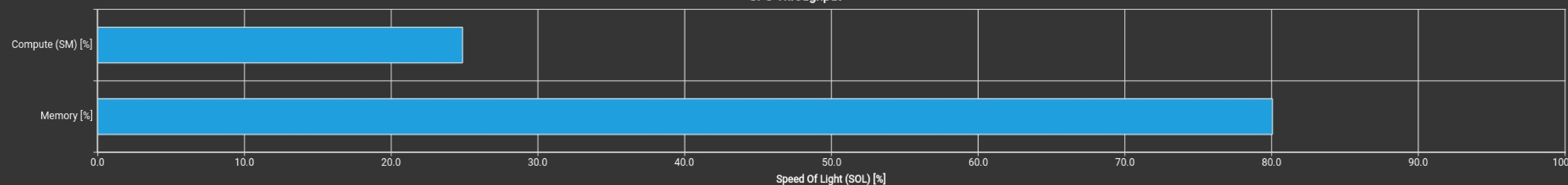
GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

Compute (SM) Throughput [%]	24.86	Duration [ms]	85.44
Memory Throughput [%]	80.06	Elapsed Cycles [cycle]	164,052,065
L1/TEX Cache Throughput [%]	87.21	SM Active Cycles [cycle]	150,343,756.88
L2 Cache Throughput [%]	10.49	SM Frequency [Ghz]	1.92
DRAM Throughput [%]	1.28	DRAM Frequency [Ghz]	7.99

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory performance of the device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing L1 in the Memory Workload Analysis section.

GPU Throughput



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	128	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	40	Static Shared Memory Per Block [byte/block]	0
Block Size	32	Dynamic Shared Memory Per Block [Kbyte/block]	8.45
Threads [thread]	4,096	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	0.53	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all	-	-

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	20.83	Block Limit Registers [block]	48
Theoretical Active Warps per SM [warp]	10	Block Limit Shared Mem [block]	10
Achieved Occupancy [%]	10.98	Block Limit Warps [block]	48
Achieved Active Warps Per SM [warp]	5.27	Block Limit SM [block]	24

Theoretical Occupancy
Est. Local Speedup: 79.17%

The 2.50 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (20.8%) is limited by the required amount of shared memory.

Key Performance Indicators

Metric Name	Value	Guidance
smsp__maximum_warps_avg_per_active_cycle	2.5	Increase the theoretical number of warps per scheduler that can be issued

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	150,343,756.88	Average L1 Active Cycles [cycle]	150,343,756.88
Average L2 Active Cycles [cycle]	69,586,870.25	Average SMSP Active Cycles [cycle]	147,588,733.39
Average DRAM Active Cycles [cycle]	8,720,636	Average MC Channel Active Cycles [cycle]	n/a
Total SM Elapsed Cycles [cycle]	3,930,581,440	Total L1 Elapsed Cycles [cycle]	3,930,581,440
Total L2 Elapsed Cycles [cycle]	2,399,253,568	Total SMSP Elapsed Cycles [cycle]	15,722,325,760
Total DRAM Elapsed Cycles [cycle]	2,731,810,816	Total MC Channel Elapsed Cycles [cycle]	n/a

SMs Workload Imbalance
Est. Speedup: 7.54%

One or more SMs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 8.22% above the average, while the minimum instance value is 4.62% below the average.

Key Performance Indicators

Metric Name	Value	Guidance
sm__cycles_active.avg	1.50344E+08	Balancing the number of active cycles across SMs would result in a more optimized workload

SMSPs Workload Imbalance
Est. Speedup: 8.92%

One or more SMSPs have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 9.90% above the average, while the minimum instance value is 5.56% below the average.

Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 7.54%

One or more L1 Slices have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 8.22% above the average, while the minimum instance value is 4.62% below the average.

Key Performance Indicators

Missing Roofline and Memory Charts? Profile again with the *detailed* metric set to collect all necessary metrics.
Also consider collecting all available sections with the *full* metric set.